

Chapter 3

Lecture 3: Continuous maps in metric spaces

Definition 3.0.1. $x \in M$ is an interior point of M if there exists $\epsilon > 0$ such that $U_\epsilon(x) \subseteq M$. (We denote $\text{Int}(M)$ as the set of all interior points of M .)

Definition 3.0.2. M is open if $M = \text{Int}(M)$.

Theorem 3.0.1. $U \subseteq \mathbb{R}$ is open $\iff U$ is a union of at most countably many disjoint open intervals.

Example 3.0.1 (discrete metric). Let (X, d) be a metric space where

$$d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

for all $x, y \in X$. Then every subset of X is open (e.g. a set of a single point, its neighborhood has radius smaller than 1).

Remark In $\mathbb{R}^n$, any "reasonable" set of strict inequalities defines an open set. For example, $\{x \in \mathbb{R}^n | x_1 > 0\}$ or $\{(x, y) \in \mathbb{R}^2 | 2x > y, y^2 + y > 3x^2 - 2\}$ are open. Another example: $X = \mathbb{R}^{n^2} \simeq \text{Mat}(n)$. Then the set $\{A \subseteq \text{Mat}(n) | \det(A) \neq 0\}$ is open where A is non-degenerate. Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad \det(A) \neq 0 \quad (3.1)$$

Metric on $\text{Mat}(n)$ is defined: $\|A\| = \max(|a_{ij}|)$. Then we have

$$\begin{aligned} U_\epsilon(A) &= \{B \in \text{Mat}(n) | \|A - B\| < \epsilon\} \\ &= \{B \in \text{Mat}(n) | \max(|a_{ij} - b_{ij}|) < \epsilon\} \\ &= \left\{ \begin{pmatrix} a_{11} + \epsilon_{11} & a_{12} + \epsilon_{12} & \cdots & a_{1n} + \epsilon_{1n} \\ a_{21} + \epsilon_{21} & a_{22} + \epsilon_{22} & \cdots & a_{2n} + \epsilon_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + \epsilon_{n1} & a_{n2} + \epsilon_{n2} & \cdots & a_{nn} + \epsilon_{nn} \end{pmatrix} \mid \epsilon_{ij} < \epsilon \right\} \end{aligned} \quad (3.2)$$

Then we have

$$\det(B) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n (a_{i\sigma(i)} + \epsilon_{i\sigma(i)}) \quad (3.3)$$

Let ϵ be small enough such that $\epsilon < \|A\|$. Then We can estimate one summand of $\det(B)$ as follows:

$$|(a_{11} + \epsilon_{11})(a_{12} + \epsilon_{12}) \cdots (a_{1n} + \epsilon_{1n}) - a_{11}a_{12} \cdots a_{1n}| < \|A\|^{n-1} \cdot (2^n - 1)\epsilon \quad (3.4)$$

Sum up all estimates, we have

$$|\det(A) - \det(B)| < n! \|A\|^{n-1} \cdot (2^n - 1)\epsilon \quad (3.5)$$

Taking ϵ small enough such that

$$n! \|A\|^{n-1} \cdot (2^n - 1)\epsilon < |\det(A)| \quad (3.6)$$

ensures that $\det(B) > 0$. Thus, $U_\epsilon(A) \subseteq \{A \in \text{Mat}(n) | \det(A) \neq 0\}$. So $\{A \in \text{Mat}(n) | \det(A) \neq 0\}$ is open. ■

Definition 3.0.3. A point $x \in X$ (not necessarily in M) is a limit point of M if for every $\epsilon > 0$, the punctured neighborhood $U_\epsilon^*(x)$ contains a point of M ($U_\epsilon^*(x) \cap M \neq \emptyset$) or (the neighborhood $U_\epsilon(x)$ contains infinitely many points of M).

Definition 3.0.4. $x \in M$ is isolated if $\exists \epsilon > 0$ such that $U_\epsilon(x) \cap M = \{x\}$.

Definition 3.0.5. The closure of M , denoted by $\overline{M}$, is the union of the set of all limit points of M and M itself.

Definition 3.0.6. A set M is closed if M contains all its limit points.

Property 3.0.1. $M = \overline{M} \iff M$ is closed

Exercise Show that for any $M \subseteq X$, we have $\overline{\overline{M}} = \overline{M}$.

Property 3.0.2. If M is open in X , then $C M = X \setminus M$ is closed in X . Conversely, if M is closed in X , then $C M$ is open in X .

Exercise Prove the above property.

Property 3.0.3. $M \subseteq X$ is closed $\iff \forall y \in X \setminus M, \exists \epsilon > 0$ such that $U_\epsilon(y) \cap M = \emptyset$. (any point can be separated from M).

Proof Suppose $\exists y \in X \setminus M$ that cannot be separated from M . Then $\forall \epsilon > 0, U_\epsilon(y) \cap M \neq \emptyset$. But $y \notin M$, so $U_\epsilon^*(y) \cap M \neq \emptyset$. So y is a limit point of M which implies that $y \in \overline{M}$. Since M is closed, we have a contradiction.

Suppose every $y \in X \setminus M$ can be separated from M . Let x be a limit point of M . If $x \notin M$, then by assumption, there exists $\epsilon > 0$ such that $U_\epsilon(x) \cap M = \emptyset$. This contradicts the definition of limit point.

Example 3.0.2. We will see that $\{\det(A) = 0\}$ is closed in $\text{Mat}(n)$. So every matrix with $\det(A) \neq 0$ can be separated from the set $\{\det(A) = 0\}$.

Definition 3.0.7. The function $f : X_1 \rightarrow X_2$ between two metric spaces $(X_1, d_1), (X_2, d_2)$ is continuous if for every $x, \forall \epsilon > 0, \exists \delta > 0$ such that $\forall x' \in X_1$ with $d_1(x, x') < \delta$, we have $d_2(f(x), f(x')) < \epsilon$.

Definition 3.0.8 (Alternative definition of continuity). The function $f : X_1 \rightarrow X_2$ between two metric spaces $(X_1, d_1), (X_2, d_2)$ is continuous if for every open set $U \subseteq X_2$, the preimage $f^{-1}(U) = \{x \in X_1 | f(x) \in U\}$ is open in X_1 .

Theorem 3.0.2. The two definitions of continuity are equivalent.

Proof Let $f : X_1 \rightarrow X_2$ be continuous in the first definition. Let $U \subseteq X_2$ be open. Let $x \in f^{-1}(U)$. Then $f(x) \in U$. Since U is open, there exists $\epsilon > 0$ such that $U_\epsilon(f(x)) \subseteq U$. By continuity of f , there exists $\delta > 0$ such that $\forall x' \in X_1$ with $d_1(x, x') < \delta$, we have $d_2(f(x), f(x')) < \epsilon$, i.e. $f(x') \in U_\epsilon(f(x)) \subseteq U$. Thus, $x' \in f^{-1}(U)$. So we have $U_\delta(x) \subseteq f^{-1}(U)$. Thus, $f^{-1}(U)$ is open in X_1 .

Let $f : X_1 \rightarrow X_2$ be continuous in the second definition. Let $x \in X_1$. Let $\epsilon > 0$. Consider the open set $U_\epsilon(f(x)) \subseteq X_2$. By continuity of f , the preimage $f^{-1}(U_\epsilon(f(x)))$ is open in X_1 . Since $x \in f^{-1}(U_\epsilon(f(x)))$, there exists $\delta > 0$ such that $U_\delta(x) \subseteq f^{-1}(U_\epsilon(f(x)))$. Thus, for every $x' \in X_1$ with $d_1(x, x') < \delta$, we have $f(x') \in U_\epsilon(f(x))$, i.e. $d_2(f(x), f(x')) < \epsilon$. Thus, f is continuous in the first definition.

Question Suppose $f : X \rightarrow Y$ is continuous and bijective, is the inverse $f^{-1} : Y \rightarrow X$ also continuous?

Answer Not necessarily. For example, let $X = \{0\} \cup (1, 2]$ and $Y = [0, 1]$. Define $f : X \rightarrow Y$ by

$$f(x) = \begin{cases} 0, & x = 0 \\ x - 1, & x \in (1, 2] \end{cases}$$

Then f is continuous and bijective. However, $\{0\} \subseteq X$ is open but its preimage $f^{-1}(\{0\}) = \{0\}$ is not open in X .

There's another example: Let $X = [0, 1]$ with discrete metric (i.e., $d(x, y) = 1, \forall x, y \in X$ with $x \neq y$) and $Y = [0, 1]$ with standard metric. Define $f : X \rightarrow Y$ by $f(x) = x$. Then f is continuous and bijective. But its inverse is not continuous.